

LUCIDPREP

Grade 9 Mathematics Curriculum

U.S. Standards-Aligned Course Syllabus

MATHEMATICS

Algebra 1

36 weeks • 4 modules • 70% mastery

Curriculum Architecture

This course is organized as 36 weeks across four 9-week modules. Every week follows LucidPrep's established secondary learning cycle: AI Instructor opening, objectives, vocabulary, five warm-ups, main lecture, two short tutorial videos, AI Instructor presentation, two homework sets, and a 40-question assessment.

Mastery standard: 70%+. Students below mastery review the instructional presentation and may retake the assessment, matching the existing LucidPrep secondary model.

U.S. Alignment Note

The U.S. does not have one national Grade 9 mathematics sequence. In most districts Grade 9 mathematics is Algebra 1. This curriculum uses the Common Core State Standards for Mathematics as the primary national reference and cross-checks major themes against California Common Core and Texas Algebra I TEKS. Course placement varies by district, so this is presented as a U.S. Standards-Aligned Grade 9 course.

Prepared for LucidPrep • Curriculum Planning Edition • 2026

Common Instructional & Assessment Model

Weekly Learning Cycle

Step	Component	Description
1	Opening / Domain Overview	~2-minute AI Instructor introduction
2	Objectives + Vocabulary	Student-facing outcomes and core terminology
3	Warm-Up	5 questions with explanations
4	Main Lecture	Concept instruction, models and examples
5	Video Tutorials	2 short instructional videos
6	AI Instructor Presentation	Visual models, examples, strategies and applications
7	Homework	Two sets of 5 questions
8	Weekly Assessment	40 questions; 70% mastery
9	Review / Retake	Below mastery → review presentation → retake

Assessment Philosophy

Questions should emphasize reasoning, explanation, modeling, interpretation of data or text, application to unfamiliar problems, and evidence-based thinking rather than recall alone. Each module assessment should sample all major concepts and include scenario, data/text and constructed-response items.

This 36-week course is designed around **CCSS Algebra 1: N-Q, N-RN, A-SSE, A-APR, A-CED, A-REI, F-IF, F-BF, F-LE, S-ID**. It is a full-year curriculum with four nine-week modules and a consistent LucidPrep mastery workflow.

Grade	9
Course	Algebra 1
Duration	36 weeks
Structure	4 modules × 9 weeks
Weekly assessment	40 questions
Mastery threshold	70%
Module assessments	Weeks 9, 18, 27
Final assessment	Week 36

Course Learning Outcomes

- Reason quantitatively and use algebraic structure to represent and solve problems.
- Model relationships with linear, exponential and quadratic functions.
- Solve and justify equations, inequalities and systems.
- Analyze functions using tables, graphs, symbols and context.
- Interpret and communicate quantitative data and statistical relationships.

Mathematics

Module 1 — Foundations, Expressions & Linear Equations

Week	Topics	Students will be able to	Standards
Week 1 — Quantitative Reasoning & Units	Real number system; quantities; units; precision; order of operations; evaluating expressions	Reason with units; evaluate and interpret expressions in context	N-Q; N-RN
Week 2 — Algebraic Expressions & Properties	Variables; terms; like terms; distributive property; properties of real numbers; simplifying	Simplify and rewrite expressions; justify each step	A-SSE.1/2
Week 3 — Solving Linear Equations	One/two-step; multi-step; variables on both sides; literal equations and formulas	Solve linear equations; rearrange to solve for a variable	A-REI.1/3; A-CED.4
Week 4 — Proportional Reasoning & Applications	Ratios; proportions; percent; rates; dimensional analysis; mixture/word problems	Model and solve proportion and percent problems	N-Q; A-CED.1
Week 5 — Linear Inequalities in One Variable	Solving inequalities; compound inequalities; number-line graphs; applications	Solve and graph inequalities; interpret the solution set	A-REI.3; A-CED.1
Week 6 — Absolute Value Equations & Inequalities	Meaning of absolute value; solving equations; solving inequalities; graphing solutions	Solve absolute value equations and inequalities	A-REI; A-CED
Week 7 — Relations & Functions	Relations; domain/range; definition of a function; function notation; discrete vs continuous	Identify functions; use and evaluate function notation	F-IF.1/2
Week 8 — Representing & Interpreting Functions	Tables; graphs; mappings; verbal descriptions; key features in context	Translate among representations; interpret key features	F-IF.4/5/9
Week 9 — Module Assessment	Integrated review of expressions, equations, inequalities and functions	Synthesize concepts; demonstrate 70%+ mastery	Performance assessment

Module delivery notes. Emphasize multiple representations — tables, graphs, symbols and context — and require justification and interpretation, not just answers. Anchor procedures in real-world modeling and connect the linear, exponential and quadratic function families.

Mathematics

Module 2 — Linear Functions, Systems & Modeling

Week	Topics	Students will be able to	Standards
Week 1 — Slope & Rate of Change	Slope from graphs, points and tables; average rate of change; interpretation	Compute and interpret slope as a rate of change	F-IF.6; A-CED
Week 2 — Graphing Linear Functions	Slope-intercept form; intercepts; graphing lines; horizontal/vertical lines	Graph linear functions from equations	F-IF.7; A-REI.10
Week 3 — Writing Linear Equations	Slope-intercept; point-slope; standard form; from two points or point and slope	Write equations of lines in multiple forms	A-CED.2; F-LE.2
Week 4 — Parallel, Perpendicular & Modeling	Parallel/perpendicular slopes; linear models; interpreting slope and intercept in context	Relate lines by slope; build and interpret linear models	F-LE.1/5; A-CED
Week 5 — Linear Inequalities in Two Variables	Graphing half-planes; boundary lines; testing points; applications	Graph and interpret two-variable inequalities	A-REI.12
Week 6 — Systems: Graphing & Substitution	Meaning of a solution; solving by graphing; substitution method	Solve systems by graphing and substitution	A-REI.6
Week 7 — Systems: Elimination & Applications	Elimination method; special cases (no/infinite solutions); word problems	Solve systems by elimination; model real situations	A-REI.5/6; A-CED.3
Week 8 — Systems of Inequalities & Optimization	Graphing systems of inequalities; feasible regions; introduction to constraints	Graph systems; interpret feasible regions	A-REI.12; A-CED.3
Week 9 — Module Assessment	Slope, linear graphs and equations, systems of equations and inequalities	Synthesize concepts; demonstrate 70%+ mastery	Performance assessment

Module delivery notes. Emphasize multiple representations — tables, graphs, symbols and context — and require justification and interpretation, not just answers. Anchor procedures in real-world modeling and connect the linear, exponential and quadratic function families.

Module 3 — Exponents, Polynomials & Factoring

Week	Topics	Students will be able to	Standards
Week 1 — Exponent Properties	Product/quotient/power rules; zero and negative exponents; scientific notation	Apply exponent laws; use scientific notation	N-RN; A-SSE
Week 2 — Radicals & Rational Exponents	Square/cube roots; simplifying radicals; rational-exponent form	Connect radicals and rational exponents	N-RN.1/2
Week 3 — Exponential Functions	Growth and decay; $y = a \cdot b^x$; tables and graphs; comparing to linear	Model exponential growth and decay	F-LE.1/2/5; F-IF.7
Week 4 — Sequences	Arithmetic and geometric sequences; recursive and explicit forms; sequences as functions	Model, extend and represent sequences	F-IF.3; F-BF.2; F-LE.2
Week 5 — Polynomials & Operations	Monomials/polynomials; degree; adding and subtracting; multiplying	Classify polynomials and operate on them	A-APR.1
Week 6 — Factoring I: GCF & Trinomials	Greatest common factor; factoring x^2+bx+c ; difference of squares	Factor basic polynomials	A-SSE.3; A-APR
Week 7 — Factoring II: General & Special Cases	Factoring ax^2+bx+c ; perfect-square trinomials; factoring by grouping	Factor a range of polynomial expressions	A-SSE.2/3
Week 8 — Applications of Exponentials & Polynomials	Modeling contexts; interpreting parameters; rewriting to reveal structure	Apply models; interpret structure in context	A-SSE.1; F-LE.5
Week 9 — Module Assessment	Exponents, radicals, exponential functions, sequences, polynomials and factoring	Synthesize concepts; demonstrate 70%+ mastery	Performance assessment

Module delivery notes. Emphasize multiple representations — tables, graphs, symbols and context — and require justification and interpretation, not just answers. Anchor procedures in real-world modeling and connect the linear, exponential and quadratic function families.

Module 4 — Quadratic Functions, Equations & Data

Week	Topics	Students will be able to	Standards
Week 1 — Introduction to Quadratic Functions	$y = ax^2 + bx + c$; parabolas; vertex, axis of symmetry, intercepts, max/min	Identify and interpret key features of parabolas	F-IF.4/7
Week 2 — Graphing Quadratic Functions	Standard and vertex form; transformations; graphing parabolas	Graph quadratics and describe transformations	F-IF.7/8; F-BF.3
Week 3 — Solving Quadratics by Factoring	Zero product property; roots and zeros; connecting factors to x-intercepts	Solve quadratic equations by factoring	A-REI.4; A-SSE.3
Week 4 — Square Roots & Completing the Square	Square-root method; completing the square; converting to vertex form	Solve quadratics by square roots and completing the square	A-REI.4; A-SSE.3
Week 5 — Quadratic Formula & Discriminant	Quadratic formula; the discriminant; number and nature of solutions	Solve with the formula; interpret the discriminant	A-REI.4
Week 6 — Modeling with Quadratics	Projectile and area problems; choosing a model; comparing linear/exponential/quadratic	Model with quadratics; compare function families	F-LE.3; A-CED; F-IF
Week 7 — Data Analysis & Statistics	Dot plots, histograms, box plots; center and spread; comparing distributions	Represent and compare data distributions	S-ID.1/2/3
Week 8 — Bivariate Data & Correlation	Scatter plots; line of best fit; correlation vs causation; two-way tables	Fit and interpret linear models for data	S-ID.6/7/8/9
Week 9 — Final Assessment	Full-course synthesis: expressions → functions → equations → modeling → data	Solve unfamiliar problems; demonstrate mastery	Final performance assessment

Module delivery notes. Emphasize multiple representations — tables, graphs, symbols and context — and require justification and interpretation, not just answers. Anchor procedures in real-world modeling and connect the linear, exponential and quadratic function families.

Assessment & Alignment

Assessment Blueprint

Warm-ups: weekly • Homework: weekly • Weekly assessment: 40 questions • Module assessments: Weeks 9 / 18 / 27 • Final: Week 36 • Below 70%: instructional review and retake.

Standards Alignment

Primary framework	CCSS: N-Q, N-RN, A-SSE, A-APR, A-CED, A-REI, F-IF, F-BF, F-LE, S-ID
Mathematical Practices	Problem solving; reasoning & proof; modeling with mathematics; multiple representations; strategic use of tools; precision; structure; regularity in repeated reasoning
Recurring Concepts	Structure; quantity; pattern & change; equivalence; modeling; data & variability
State mapping	California Common Core and Texas TEKS can be mapped at chapter / question level without changing the core 36-week structure.

Content Production Guidance

For each week, expand the syllabus topic into the established LucidPrep package: ~2-minute opening, objectives, vocabulary, 5 warm-ups with explanations, lecture, 2 ~2-minute videos, AI Instructor presentation, two homework sets of 5, 40-question assessment, answer explanations and mastery/retake handling. Build question banks around problems, scenarios, data, texts and application.

Curriculum Basis & Source Notes

This document is a U.S.-aligned curriculum architecture, not a claim of a single nationwide Grade 9 sequence. In most U.S. districts Grade 9 mathematics is Algebra 1; course placement and titles vary by state and district.

Primary reference	Common Core State Standards for Mathematics — high-school conceptual categories (Number & Quantity, Algebra, Functions, Statistics & Probability) as realized in the Algebra 1 course.
State cross-check	California Department of Education — California Common Core State Standards for Mathematics, Algebra 1.
State cross-check	Texas Education Agency — Algebra I TEKS: linear functions, equations and inequalities; quadratic functions and equations; exponential functions and equations; and number & algebraic methods.
LucidPrep product alignment	36-week / four-module secondary architecture, weekly instructional components, 40-question assessments, 70% mastery, and review/retake pathway.

Recommended next production step. Add standards metadata to each chapter and question (CCSS cluster/standard codes plus state-framework identifiers) so LucidPrep can support multiple U.S. state alignments without duplicating the course.